

Math 101: Linear and Multilinear Algebra

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Worksheet 1.1: Rings, Homomorphisms, and Quotients

A *ring* $(R, +, \cdot)$ is an abelian group $(R, +, 0_R)$ together with a multiplication $\cdot$ and an element $1_R \in R$ satisfying the following axioms:

- (i) (Identity) $1_R \cdot a = a \cdot 1_R = a$ for all $a \in R$,
- (ii) (Associativity) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R$,
- (iii) (Left Distributivity) $a \cdot (b + c) = a \cdot b + a \cdot c$ for all $a, b, c \in R$, and
- (iv) (Right Distributivity) $(a + b) \cdot c = a \cdot c + b \cdot c$ for all $a, b, c \in R$.

We call 1_R the *multiplicative identity*. We generally write ab for $a \cdot b$, and we will write 1 and 0 for 1_R and 0_R when the underlying ring is clear from context. A ring is *commutative* if $ab = ba$ for all $a, b \in R$.

Throughout this course, all rings have a multiplicative identity, but need not be commutative. In particular, matrix rings will be just as important to us as the ring of integers, or polynomial rings. The *zero ring* is the unique ring with a single element which satisfies $1 = 0$. An element $u \in R$ is a *unit* if there is an element $v \in R$ such that $uv = vu = 1$. In this case we say that v is the *multiplicative inverse* of u , which we generally denote by u^{-1} . We write $R^\times$ for the set of units of R . A *field* is a commutative ring with $1 \neq 0$ in which every nonzero element is a unit. The letter $\mathbb{K}$ will denote a field throughout these worksheets.

The goal of this worksheet is to introduce three related ways of studying rings: their elements, their homomorphisms, and their ideals. You already know a number of examples, including $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and matrix rings. The definitions let us discuss the features these examples share, while retaining differences such as whether multiplication is commutative or whether nonzero elements can be inverted. We begin with some calculations that illustrate these differences.

Problem 1. Basic Ring Identities. Throughout this question let R be a ring.

(a) Starting from the axioms, prove the following:

- (i) If 1 and $1'$ both satisfy the identity axiom then $1 = 1'$. Hint: evaluate $1 \cdot 1'$ in two different ways.
- (ii) If 0 and $0'$ both satisfy the property of being additive identities then $0 = 0'$.
- (iii) If $a + b = 0$ and $a + c = 0$ then $b = c$.

Hence for any ring the additive and multiplicative identity elements are unique, and for any element $a \in R$ there is a unique additive inverse, which we generally denote $-a$.

(b) Starting from the axioms, prove the following identities:

- (i) $a0 = 0a = 0$ for all $a \in R$,
- (ii) $(-a)b = -(ab) = a(-b)$ for all $a, b \in R$.

(c) If $1 = 0$, explain why the ring must have only one element.

(d) Suppose $vu = 1$ and $uw = 1$. Compute $v(uw)$ and $(vu)w$ and conclude $v = w$. Deduce that u is a unit and that its inverse is unique, so that the notation u^{-1} is justified. Deduce also that $(u^{-1})^{-1} = u$.

(e) Prove that $R^\times$ is a group under multiplication, with $(uv)^{-1} = v^{-1}u^{-1}$.

Solution.

- (a) (i)
- (ii)
- (iii)
- (b) (i)
- (ii)
- (c)
- (d)
- (e)

Problem 2. First Examples of Rings.

- (a) For a positive integer n , use the familiar additive quotient group $\mathbb{Z}/n\mathbb{Z}$ and define $\overline{a}\overline{b} = \overline{ab}$. Check that this does not depend on the chosen representatives. What is the identity?
- (b) Characterize the units in the following rings:
- | | | |
|--------------------|--------------------------------|---------------------------------|
| (i) $\mathbb{Z}$ | (iv) $\mathbb{K}$ | (vii) $\mathbb{Z}/4\mathbb{Z}$ |
| (ii) $\mathbb{C}$ | (v) $\mathbb{Z}/12\mathbb{Z}$ | (viii) $\mathbb{Z}/2\mathbb{Z}$ |
| (iii) $\mathbb{R}$ | (vi) $\mathbb{Z}/30\mathbb{Z}$ | (ix) $\mathbb{Z}/n\mathbb{Z}$ |
- (c) A nonzero element $a \in R$ is a *zero-divisor* if there exists a nonzero element $b \in R$ such that $ab = ba = 0$. Characterize the zero divisors in all of the rings from part (b).
- (d) Let $\text{Mat}_n(\mathbb{K})$ denote the set of $n \times n$ matrices over $\mathbb{K}$. Prove that $\text{Mat}_n(\mathbb{K})$ is a ring under normal matrix addition and multiplication. What are the identity elements?
- (e) For integers $1 \leq i, j \leq n$ let $E_{i,j}$ denote the $n \times n$ matrix whose (i, j) -th entry is 1 and all other entries are zero. Compute $E_{12}E_{21}$ and $E_{21}E_{12}$. What does this show about the ring $\text{Mat}_n(\mathbb{K})$?
- (f) Explain why the units of $\text{Mat}_2(\mathbb{K})$ are precisely the invertible matrices. Compare “nonzero” and “invertible” in $\text{Mat}_2(\mathbb{K})$ and in $\mathbb{K}$.
- (g) Give an example of two elements $u, v \in \text{Mat}_2(\mathbb{R})$ showing that in general $(uv)^{-1} = u^{-1}v^{-1}$ can fail.
- (h) Let R be a ring and $\text{Mat}_n(R)$ denote the set of $n \times n$ matrices whose entries are elements in R . Define a notion of matrix multiplication and matrix addition for elements of $\text{Mat}_n(R)$ that agrees with our normal matrix operations when $R = \mathbb{K}$. Is $\text{Mat}_n(R)$ a ring?

Solution.

(a)

(b) (i)

(ii)

(iii)

- (iv)
- (v)
- (vi)
- (vii)
- (viii)
- (ix)
- (c)
- (d)
- (e)
- (f)
- (g)
- (h)

If R is a ring the *polynomial ring* with coefficients in R is the set $R[x]$ consisting of finite formal sums $a_0 + a_1x + \cdots + a_dx^d$ with coefficients in R . Addition is coefficient-wise, and multiplication is obtained by distributing, using $x^ix^j = x^{i+j}$, and collecting coefficients of the same power of x . We can regard elements of R as constant polynomials: their addition and multiplication in $R[x]$ agree with the original operations on R . The word “formal” means that equality is determined by the coefficients. This is slightly different from thinking of a polynomial as a function on R . Given an element $a \in R$, we can substitute a for x to obtain a value $f(a) \in R$, but the polynomial f and the value $f(a)$ are different objects. Over some fields, even two different polynomials can define the same function. We will see an example below and then return to substitution when we discuss ring homomorphisms.

For a nonzero polynomial f , its *degree*, denoted $\deg(f)$, is the largest exponent with nonzero coefficient. That coefficient is called the *leading coefficient*. Looking at leading coefficients is often a useful way to understand multiplication without expanding every term of a product.

Problem 3. Polynomial Rings.

- (a) Compute $(a + bx)(c + dx)$ in $\mathbb{K}[x]$. Identify the coefficient of x^j in the product of two general polynomials.
- (b) By looking at leading coefficients, prove $\deg(fg) = \deg(f) + \deg(g)$ for nonzero $f, g \in \mathbb{K}[x]$. Explain why the same argument works in $\mathbb{Z}[x]$.
- (c) Determine the units of $\mathbb{K}[x]$. In particular, can you divide by x in this ring?
- (d) Let $\mathcal{F}_2 = \mathbb{Z}/2\mathbb{Z}$. Evaluate $x^2 - x$ at both elements of $\mathcal{F}_2$. Is $x^2 - x$ the zero polynomial? Explain the difference between equality of polynomials and equality of the functions they define on a field.

Solution.

- (a)
- (b)
- (c)
- (d)

A *ring homomorphism* is a function $\phi : R \rightarrow S$ satisfying the following conditions for all $a, b \in R$:

- (i) $\phi(a + b) = \phi(a) + \phi(b)$.
- (ii) $\phi(ab) = \phi(a)\phi(b)$.
- (iii) $\phi(1_R) = 1_S$.

The first condition says that a ring homomorphism is a homomorphism of the underlying additive groups. In particular, a ring homomorphism preserves 0 and additive inverses. The second condition ensures that a ring homomorphism respects the multiplicative structure of both rings. Finally, the last condition ensures that all ring homomorphisms preserve the multiplicative identity. Condition (iii) does not follow from the others, and some others do not make this assumption in their definition of ring homomorphisms.

A ring homomorphism $\phi : R \rightarrow S$ is a *ring isomorphism* if it has an inverse $\phi^{-1} : S \rightarrow R$, which is also a ring homomorphism. Just as with groups, the condition of a ring homomorphism being a ring isomorphism is essentially a set theoretic one: a ring homomorphism $\phi : R \rightarrow S$ is an isomorphism if and only if ϕ is bijective. You will check this equivalence shortly. We say that two rings R and S are isomorphic, and write $R \cong S$, if there exists an isomorphism between them. An isomorphism allows us to identify two rings which are essentially the same up to relabeling their elements.

As with group homomorphisms, a ring homomorphism comes with two very important subsets; its *kernel* and its *image*:

$$\ker(\phi) := \{a \in R \mid \phi(a) = 0_S\} \subset R \quad \text{and} \quad \text{img}(\phi) := \{\phi(a) \mid a \in R\} \subset S.$$

Notice that a map is surjective exactly when its image is the whole target, while its kernel measures its failure to be injective.

A *subring* of R is a subset $T \subset R$ that is a ring under the inherited operations and contains 1_R . The image of a ring homomorphism is a subring of its target. The kernel has a different closure property, which will lead us shortly to the notion of ideals.

Problem 4. Examples and Properties of Ring Homomorphisms. Throughout this exercise let R be a ring.

- (a) Prove that a ring homomorphism preserves 0, negatives, and units. If u is a unit, what must $\phi(u)^{-1}$ be?
- (b) Show that there is exactly one ring homomorphism $\phi : \mathbb{Z} \rightarrow R$. Write down a concrete description of this homomorphism when $R = \mathbb{Z}, \mathbb{Q}, \mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/12\mathbb{Z}$, and $\text{Mat}_2(\mathbb{R})$.

- (c) For each of the following examples determine whether a ring homomorphism exists, and if it does, whether it is unique:

$$\mathbb{Z}/6\mathbb{Z} \longrightarrow \mathbb{Z}/3\mathbb{Z}, \quad \mathbb{Z}/3\mathbb{Z} \longrightarrow \mathbb{Z}/6\mathbb{Z}, \quad \mathbb{Z}/4\mathbb{Z} \longrightarrow \mathbb{Z}/4\mathbb{Z}.$$

Hint: Start by deciding where $\bar{1}$ would have to go, and then use an additive relation it satisfies.

- (d) For $a \in R$, the *evaluation at a homomorphism* is given by $\text{ev}_a : R[x] \rightarrow R$ by $f \mapsto f(a)$. Carefully check that ev_a is a surjective ring homomorphism.
- (e) Let $\phi : R \rightarrow S$ be a ring homomorphism. Prove that $\text{img}(\phi) \subset S$ is a subring of S .
- (f) Let $\phi : R \rightarrow S$ be a ring homomorphism. Prove the following:
- (i) Prove that ϕ is surjective if and only if $\text{img}(\phi) = S$.
 - (ii) Prove that ϕ is injective if and only if $\ker(\phi) = \{0\}$.
 - (iii) Prove that ϕ is an isomorphism if and only if ϕ is bijective.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)
 - (i)
 - (ii)
 - (iii)

We would now like to describe the subsets of a ring that can occur as kernels. The kernel of a ring homomorphism is an additive subgroup, but it satisfies more than closure under addition. If $\phi(a) = 0$, then $\phi(ra) = \phi(r)\phi(a) = 0$ and $\phi(ar) = \phi(a)\phi(r) = 0$ for every $r \in R$. Thus multiplying an element of the kernel by an arbitrary ring element, on either side, keeps us in the kernel. A *two-sided ideal* $I \subset R$ is an additive subgroup such that $ra \in I$ and $ar \in I$ for all $r \in R$ and $a \in I$. A *left ideal* requires only the condition $ra \in I$. A *right ideal* requires only that $ar \in I$.

Unless we say otherwise, *ideal* will always mean two-sided ideal. In a commutative ring this distinction disappears, however, in arbitrary rings it is very important. For example, we will see an example of a left ideal in a matrix ring that is not a two-sided ideal. Note also that an ideal need not be a subring under our convention: a proper ideal does not contain 1_R .

For elements $a_1, \dots, a_t$ in a commutative ring R , we write

$$\langle a_1, \dots, a_t \rangle := \{r_1 a_1 + \dots + r_t a_t \mid r_i \in R\},$$

which is the ideal *generated* by $a_1, \dots, a_t$. More generally, if we do not assume R is commutative the ideal *generated* by elements $a_1, \dots, a_t \in R$ is

$$\langle a_1, \dots, a_t \rangle := \{r_1 a_1 s_1 + \dots + r_t a_t s_t \mid r_i, s_i \in R\}.$$

It is worth thinking for a moment, why these two definitions agree. In both situations we say that an ideal $I \subset R$ is a *principal ideal* if and only if there exists $a \in R$ such that $I = \langle a \rangle$, i.e., I is generated by a single element. More intrinsically, the ideal generated by $a_1, \dots, a_t \in R$ is the smallest ideal of R containing all of the a_i simultaneously.

Problem 5. Properties & Examples of Ideals. Throughout this exercise let R be a ring.

- (a) Let $I, J \subset R$ be ideals. Prove that $I \cap J$ is an ideal.
- (b) Give an example to show that if $I, J \subset R$ are ideals then $I \cup J$ need not be an ideal.
- (c) Let $I, J \subset R$ be ideals. Prove that the *ideal sum* of I and J defined below is an ideal:

$$I + J := \{a + b \mid a \in I, b \in J\}.$$

- (d) Let $a_1, \dots, a_t \in R$. Prove $\langle a_1, \dots, a_t \rangle \subset R$ is a two-sided ideal.
- (e) Let $a_1, \dots, a_t \in R$. Prove that $\langle a_1, \dots, a_t \rangle$ is the smallest ideal of R containing

$a_1, \dots, a_t$, i.e.,

$$\langle a_1, \dots, a_t \rangle = \bigcap_{\substack{J \subset R \\ a_1, \dots, a_t \in J}} J.$$

- (f) Let $I \subset R$ be an ideal. Prove that $I = R$ if and only if I contains a unit. Deduce that the only ideals of a field $\mathbb{K}$ are $\{0\}$ and $\mathbb{K}$ itself.
- (g) Explicitly describe the ideals $\langle n \rangle \subset \mathbb{Z}$ and $\langle x^2 \rangle \subset \mathbb{K}[x]$. Which polynomials occur in the second example, and is x one of them?
- (h) Let $L \subset \text{Mat}_2(\mathbb{K})$ denote the set of matrices whose second column is zero. Prove that L is a left ideal. Compute $E_{11}E_{12}$ and use it to show that L is not a right ideal, and hence not a two-sided ideal.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)
- (g)
- (h)

Problem 6. Ideals and Homomorphisms. Throughout this exercise let $\phi : R \rightarrow S$ be a ring homomorphism.

- (a) Prove that $\ker(\phi)$ is a two-sided ideal of R .
- (b) Prove that if $J \subset S$ is an ideal then $\phi^{-1}(J)$ is an ideal. Explain why part (a) is the special case when $J = \langle 0 \rangle$.
- (c) Consider the inclusion $\psi : \mathbb{Z} \hookrightarrow \mathbb{Z}[x]$ given by $1 \mapsto 1$ and the ideal $\langle 2 \rangle \subset \mathbb{Z}$. Show that $\psi(\langle 2 \rangle) \subset \mathbb{Z}[x]$ is not an ideal. Hence the image of an ideal need not be an ideal.
- (d) Prove that if ϕ is surjective then $\phi(I) \subset S$ is an ideal for all ideals $I \subset R$.

Solution.

- (a)
- (b)
- (c)
- (d)

Let $I \subset R$ be a two-sided ideal. Since I is an additive subgroup and the additive group of R is abelian, we can form the quotient group R/I . Its elements are cosets $r + I$, and $r + I = s + I$ if and only if $r - s \in I$. Addition is given by $(r + I) + (s + I) = (r + s) + I$. To obtain a *quotient ring*, we define multiplication by

$$(r + I)(s + I) = rs + I.$$

The fact that I is a two-sided ideal guarantees that this formula is well defined. Together with the identity $1_R + I$, it makes R/I a ring! From group theory we know that a quotient group R/I comes with a special group homomorphism we call the *quotient map*

$$R \xrightarrow{\pi} R/I \qquad r \longmapsto r + I.$$

The map π is surjective and the kernel of π , as a group homomorphism, is precisely I . However, multiplication on R/I was defined precisely so that $\pi(r)\pi(s) = (r + I)(s + I) = rs + I = \pi(rs)$, and $\pi(1_R) = 1_R + I$. Hence π is actually a surjective ring homomorphism with kernel I . Notice that this really came down to the right definition of R/I . Consequently, every ideal occurs as the kernel of a ring homomorphism, and by Problem these are the only subsets which do.

One way to think of R/I is as the ring obtained when we impose the equations $a = 0$ for all $a \in I$. For example, in $\mathbb{Z}/n\mathbb{Z}$ we are imposing the equation $n = 0$, which forces integers differing by a multiple of n to represent the same element. More generally, a ring homomorphism $\phi : R \rightarrow S$ which sends every element of I to zero determines a unique ring homomorphism $\bar{\phi} : R/I \rightarrow S$ such that $\bar{\phi} \circ \pi = \phi$ where π is the quotient homomorphism. We often express this relationship by drawing a *commutative diagram*

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ \pi \downarrow & \nearrow \bar{\phi} & \\ R/I & & \end{array}$$

In particular, homomorphisms out of R/I can be described entirely in terms of homomorphisms out of R , namely those which kill I . This is what is often called the universal property of quotient rings. We will prove the construction and this property together in the following exercise, and return to the general language of universal properties next week.

Problem 7. Construction and Properties of Quotient Rings. Throughout this exercise let I be a two-sided ideal of a ring R .

- (a) Suppose $r' = r + a$ and $s' = s + b$ with $a, b \in I$. Expand $r's' - rs$ and prove that

it lies in I . Conclude that the proposed multiplication on R/I is well defined.

- (b) Write down the formulas for addition, multiplication, and the identity element in R/I . Verify associativity and both distributive laws by computing with representatives. Why is R/I still a ring when $I = R$, and which ring is it?
- (c) Prove that $\pi : R \rightarrow R/I, r \mapsto r + I$, is a surjective ring homomorphism with $\ker(\pi) = I$.
- (d) Return to the left ideal $L \subset \text{Mat}_2(\mathbb{K})$ from Problem . We have $E_{11} + L = 0 + L$. Multiply these two representatives on the right by E_{12} and compare the resulting cosets. Explain why the proposed multiplication on $\text{Mat}_2(\mathbb{K})/L$ is not well defined, and which hypothesis of part (a) has been lost.
- (e) Suppose $\phi : R \rightarrow S$ is a ring homomorphism with $I \subset \ker(\phi)$. Prove that $\bar{\phi}(r + I) = \phi(r)$ is a well-defined ring homomorphism $R/I \rightarrow S$, and that it is the unique one satisfying $\bar{\phi} \circ \pi = \phi$. Explain in words why a homomorphism out of R/I is the same thing as a homomorphism out of R which sends I to zero.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)

The construction of quotient rings also clarifies the relationship between the kernel and the image of a ring homomorphism. A ring homomorphism $\phi : R \rightarrow S$ respects the relation $r + I = r' + I$ precisely when it sends every element of I to zero. Whenever $I \subseteq \ker(\phi)$, we can therefore define a ring homomorphism

$$\bar{\phi} : R/I \longrightarrow S, \quad r + I \longmapsto \phi(r).$$

Thus ϕ can be computed in two steps: first pass to the quotient, and then apply $\bar{\phi}$. Taking $I = \ker(\phi)$ gives a particularly useful description. Two elements of R have the same image exactly when their difference lies in $\ker(\phi)$, so passing to this quotient identifies exactly the elements that ϕ identifies. The induced map is therefore injective, and its image is $\text{img}(\phi)$.

Theorem 0.1 (First Isomorphism Theorem for Rings). *Let $\phi : R \rightarrow S$ be a ring homomorphism. Then the map*

$$\bar{\phi} : R/\ker(\phi) \longrightarrow \text{img}(\phi), \quad r + \ker(\phi) \longmapsto \phi(r),$$

is an isomorphism of rings. In particular, there is a commutative diagram

$$\begin{array}{ccc} R & \xrightarrow{\phi} & S \\ \pi \downarrow & & \uparrow \\ R/\ker(\phi) & \xrightarrow[\sim]{\bar{\phi}} & \text{img}(\phi) \end{array}$$

Every ring homomorphism can therefore be understood as a quotient map followed by an isomorphism and an inclusion. If ϕ is surjective, then $\text{img}(\phi) = S$, and the theorem gives $R/\ker(\phi) \cong S$. This provides our standard method for recognizing a quotient ring: construct a surjective homomorphism onto a ring we already understand, and then compute its kernel.

Problem 8. First Isomorphism Theorem and Evaluation Maps.

- (a) Use Problem 1 to construct the displayed map. Why is $\text{img}(\phi)$ a ring, and why is its multiplicative identity the correct one?
- (b) Prove that the displayed map is surjective onto $\text{img}(\phi)$, and that if two cosets have the same image then they are equal. Combine this with Problem 2 to finish the proof of the theorem.
- (c) Apply the theorem to the reduction homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$. Identify the kernel and describe the induced isomorphism explicitly. What does it do to the coset of 1?

- (d) For $a \in \mathbb{K}$, prove that $\ker(\text{ev}_a) = \langle x - a \rangle$. One inclusion is immediate. For the other, first factor $x^j - a^j$ as $(x - a)(x^{j-1} + x^{j-2}a + \cdots + a^{j-1})$ and then apply this to $f(x) - f(a)$.
- (e) Deduce that $\mathbb{K}[x] / \langle x - a \rangle \cong \mathbb{K}$, and write the isomorphism down explicitly. What does the class of x become? Explain why this quotient is a genuinely different construction from the polynomial ring $\mathbb{K}[x]$ itself.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)

For the evaluation homomorphism $\text{ev}_a : \mathbb{K}[x] \rightarrow \mathbb{K}$, the kernel $\langle x - a \rangle$ imposes the relation $x = a$, and the resulting quotient is therefore isomorphic to $\mathbb{K}$. We can use the same construction to impose a polynomial relation with a rather different effect. Consider the ring:

$$\mathbb{K}[\varepsilon] := \mathbb{K}[x] / \langle x^2 \rangle$$

where $\varepsilon = x + \langle x^2 \rangle$. This ring is called the ring of *dual numbers* over $\mathbb{K}$. The relation imposed by the quotient is $\varepsilon^2 = 0$. It does not impose $\varepsilon = 0$, since the polynomial x is not a multiple of x^2 . The distinction is what makes the example useful: it gives us a very simple commutative ring containing a nonzero element whose square is zero.

An element $a \in R$ of a ring is *nilpotent* if $a^m = 0$ for some integer $m \geq 1$. Recall from Problem that a nonzero element a is a zero divisor if $ab = ba = 0$ for some nonzero $b \in R$. Hence a nonzero nilpotent element is always a zero divisor, however, the converse is not true. Neither phenomenon can occur in a field, as you saw when you computed the units of $\mathbb{K}$.

Every element of $\mathbb{K}[\varepsilon]$ can be expressed using only the classes of 1 and x , because every higher power of x already lies in $\langle x^2 \rangle$. In the following exercise we will show that this expression is unique, use it to describe the operations in $\mathbb{K}[\varepsilon]$, and determine which elements are units. We will revisit the same example as a module in the next worksheet.

Problem 9. Computations with the Dual Numbers. Throughout this exercise let $\mathbb{K}[\varepsilon] = \mathbb{K}[x] / \langle x^2 \rangle$ and $\varepsilon = x + \langle x^2 \rangle$.

- Prove that every element of $\mathbb{K}[\varepsilon]$ can be written uniquely as $a + b\varepsilon$ with $a, b \in \mathbb{K}$. Hint: for uniqueness, use your description of $\langle x^2 \rangle$ from Problem . Deduce that $\varepsilon \neq 0$ and $\varepsilon^2 = 0$.
- Derive formulas for addition and multiplication in these coordinates. Compute $(1 + \varepsilon)^2$, and say what happens when $\mathbb{K} = \mathcal{F}_2$.
- Suppose $(a + b\varepsilon)(c + d\varepsilon) = 1$ and compare coefficients. Prove that $a + b\varepsilon$ is a unit if and only if $a \neq 0$, and compute its inverse when it exists. Describe $\mathbb{K}[\varepsilon]^\times$.
- Determine all nilpotent elements and all nonzero zero divisors of $\mathbb{K}[\varepsilon]$. For the nilpotents, begin by considering the constant coefficient of a power. For the zero divisors, use your description of $\mathbb{K}[\varepsilon]^\times$.
- Construct the homomorphism $\mathbb{K}[\varepsilon] \rightarrow \mathbb{K}$ sending $a + b\varepsilon$ to a . Compute its kernel and identify the resulting quotient using the first isomorphism theorem.
- More generally, prove that for $m \geq 2$ every class in $\mathbb{K}[x] / \langle x^m \rangle$ has a unique representative $a_0 + a_1x + \cdots + a_{m-1}x^{m-1}$. Exhibit a nonzero nilpotent ele-

ment. Which parts of your argument for $m = 2$ still work, and which need modification?

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)

So far we have used homomorphisms to compare two different rings. It is equally useful to consider the homomorphisms from a ring to itself, since these record the internal symmetries of that ring. A *ring automorphism* of R is a ring isomorphism $\phi : R \rightarrow R$, and we write $\text{Aut}(R)$ for the set of all automorphisms of R . The identity map Id_R is an automorphism, a composition of two automorphisms is again an automorphism, and the inverse of an automorphism is one by definition. Hence $\text{Aut}(R)$ is a group under composition, exactly as for groups and vector spaces. An automorphism is a relabeling of the elements of R which preserves every feature expressible in terms of $+$, $\cdot$, and 1 . Units are sent to units, nilpotents to nilpotents, zero divisors to zero divisors, and ideals to ideals.

Every ring comes with one systematic supply of automorphisms. The *center* of a ring R is

$$Z(R) := \{z \in R \mid zr = rz \text{ for all } r \in R\}.$$

Notice that R is commutative if and only if $Z(R) = R$. For a unit $u \in R^\times$, conjugation $\gamma_u(r) = uru^{-1}$ is an automorphism of R , and automorphisms of this form are called *inner*. In a commutative ring every inner automorphism is the identity, so for rings such as $\mathbb{Z}$ or $\mathbb{K}[x]$ the interesting automorphisms must come from elsewhere; typically they come from substitution, as in the polynomial case where x may be replaced by $ax + b$.

Frequently we want to compare automorphisms relative to a subring which we regard as fixed. If $T \subset R$ is a subring, we write $\text{Aut}_T(R)$ for the subgroup of automorphisms which fix every element of T . Complex conjugation is an element of $\text{Aut}_{\mathbb{R}}(\mathbb{C})$, and the ring $\mathbb{K}[x]$ is most naturally studied through $\text{Aut}_{\mathbb{K}}(\mathbb{K}[x])$. This relative point of view is the starting point for Galois theory, which is a major focus of Math 111.

Problem 10. Automorphisms of Rings. Throughout this exercise let R be a ring.

- (a) Prove that $\text{Aut}(R)$ is a group under composition. Prove that any $\phi \in \text{Aut}(R)$ restricts to a group automorphism of $R^\times$, and that ϕ carries nilpotent elements to nilpotent elements and zero divisors to zero divisors.
- (b) Prove that $\phi(I)$ is an ideal for every ideal $I \subset R$ and every $\phi \in \text{Aut}(R)$, and that $I \mapsto \phi(I)$ is a bijection from the set of ideals of R to itself. Hint: use Problem .
- (c) Prove that $\text{Aut}(\mathbb{Z})$ and $\text{Aut}(\mathbb{Z}/n\mathbb{Z})$ are both trivial. Hint: use the uniqueness statement of Problem . Prove also that $\text{Aut}(\mathbb{Q})$ is trivial, by first showing that any automorphism fixes $\mathbb{Z}$ pointwise.
- (d) For $u \in R^\times$ define $\gamma_u(r) = uru^{-1}$. Prove that $\gamma_u \in \text{Aut}(R)$ and that $u \mapsto \gamma_u$ is a group homomorphism $R^\times \rightarrow \text{Aut}(R)$. Prove that its kernel is $Z(R) \cap R^\times$, and deduce that every inner automorphism is trivial precisely when R is commutative.

- (e) Let $a \in \mathbb{K}^\times$ and $b \in \mathbb{K}$. Adapting your argument from Problem , show that $f(x) \mapsto f(ax + b)$ is a ring homomorphism $\mathbb{K}[x] \rightarrow \mathbb{K}[x]$ fixing $\mathbb{K}$ pointwise, and exhibit its inverse. Conversely, suppose $\phi \in \text{Aut}_{\mathbb{K}}(\mathbb{K}[x])$. Using the degree formula of Problem , prove that $\deg(\phi(x)) = 1$, and conclude that $\text{Aut}_{\mathbb{K}}(\mathbb{K}[x])$ consists exactly of these substitutions.
- (f) Prove that complex conjugation is an automorphism of $\mathbb{C}$ fixing $\mathbb{R}$ pointwise. Prove that $\text{Aut}_{\mathbb{R}}(\mathbb{C})$ has exactly two elements. Hint: such a ϕ is determined by $\phi(i)$, and $\phi(i)^2 = -1$.
- (g) Let A be the ring of dual numbers. Prove that any $\phi \in \text{Aut}_{\mathbb{K}}(A)$ satisfies $\phi(\varepsilon) = b\varepsilon$ for some $b \in \mathbb{K}^\times$, and that every such b arises. Hint: use your classification of the nilpotents and units from Problem . Deduce that $\text{Aut}_{\mathbb{K}}(A) \cong \mathbb{K}^\times$.
- (h) Prove that $\gamma_u = \gamma_v$ in $\text{Aut}(\text{Mat}_n(\mathbb{K}))$ if and only if u and v differ by a nonzero scalar matrix, so that the inner automorphisms of $\text{Mat}_n(\mathbb{K})$ form a copy of $\text{GL}_n(\mathbb{K})/\mathbb{K}^\times$. It is a theorem of Skolem and Noether that these are in fact all of the automorphisms of $\text{Mat}_n(\mathbb{K})$; we will not prove this, but it is worth comparing with parts (e) and (g), where the automorphism groups were visibly not inner.

Solution.

- (a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)

The ideals $\langle n \rangle \subset \mathbb{Z}$ and $\langle x - a \rangle, \langle x^2 \rangle \subset \mathbb{K}[x]$ are all principal. The definition of a generated ideal allows several generators, and it is natural to ask whether they can always be replaced by a single one. This is not the case in general, even for familiar commutative rings. We will prove that the ideal $\langle 2, x \rangle \subset \mathbb{Z}[x]$ is not principal. The argument uses only the degree of a product and the constant coefficient of a polynomial, so it does not require any general theory of factorization.

Problem 11. A Nonprincipal Ideal in $\mathbb{Z}[x]$. Throughout this exercise let $I = \langle 2, x \rangle \subset \mathbb{Z}[x]$.

- (a) Prove that I consists exactly of the polynomials with even constant coefficient. For the harder inclusion, separate the constant term of a polynomial from all of the terms involving x .
- (b) Define $\theta : \mathbb{Z}[x] \rightarrow \mathcal{F}_2$ by $\theta(f) = \overline{f(0)}$. Prove that θ is a surjective ring homomorphism with $\ker(\theta) = I$. Using Problem , deduce that $\mathbb{Z}[x]/I \cong \mathcal{F}_2$ and in particular that $I \neq \mathbb{Z}[x]$.
- (c) Suppose for contradiction that $I = \langle d \rangle$ for some $d \in \mathbb{Z}[x]$. Since $2 \in I$, there is a polynomial h with $2 = dh$. Use Problem to show that d must be a nonzero constant integer.
- (d) Since $x \in I$, we may also write $x = dq$ for some $q \in \mathbb{Z}[x]$. Compare the coefficients of x on both sides. What does this force d to be?
- (e) Explain why the conclusion of (d) contradicts (b), and conclude that I is not principal. Where exactly did the argument use both of the generators?

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)

We finish by introducing several additional examples. Some extend constructions we have already met, such as polynomial rings and matrix rings; others produce a ring from a group or from a collection of operators. These examples will recur as the course develops. At this stage we will focus on their operations and on a few distinguishing properties.

- *Formal power series.* A formal power series is like a polynomial, except that it may have infinitely many nonzero coefficients. The ring $R[[t]]$ consists of expressions

$$\sum_{j \geq 0} a_j t^j = a_0 + a_1 t + a_2 t^2 + \cdots$$

for $a_j \in R$. We add and multiply using the same rules as for polynomials. For example, the product of $a_0 + a_1 t + a_2 t^2 + \cdots$ and $b_0 + b_1 t + b_2 t^2 + \cdots$ begins

$$a_0 b_0 + (a_0 b_1 + a_1 b_0) t + (a_0 b_2 + a_1 b_1 + a_2 b_0) t^2 + \cdots$$

In general, the coefficient of t^n is $\sum_{i=0}^n a_i b_{n-i}$. Although the series can be infinite, computing any one coefficient requires only finitely many operations in R . The word “formal” means that we work with the coefficients without assigning a value to t or asking whether the series converges. Two formal power series are equal exactly when all their corresponding coefficients agree.

- *Endomorphism rings.* An *endomorphism* is a homomorphism from an object to itself. If $(B, +)$ is an abelian group, the set of its endomorphisms is denoted $\text{End}_{\mathbb{Z}}(B)$. We make this set into a ring by defining

$$(f + g)(b) = f(b) + g(b), \quad (fg)(b) = f(g(b)).$$

Thus addition means adding the outputs, while multiplication means applying g first and then f . The zero element is the map sending every element to 0, and the multiplicative identity is the identity map Id_B . Similarly, the linear maps from a vector space V to itself form a ring $\text{End}_{\mathbb{K}}(V)$. If V has dimension n , choosing a basis represents these maps by $n \times n$ matrices, with addition and composition corresponding to matrix addition and multiplication.

- *Upper-triangular matrices.* A 2×2 matrix is *upper triangular* if its lower-left entry is zero. These matrices form a subring of $\text{Mat}_2(\mathbb{K})$: adding, subtracting, or multiplying two of them again gives an upper-triangular matrix, and the identity matrix is upper triangular. In particular,

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix} = \begin{pmatrix} aa' & ab' + bd' \\ 0 & dd' \end{pmatrix}.$$

This ring is not commutative. For example, if

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

then $PN = N$ but $NP = 0$.

- *Group rings.* Let G be a group. For each $g \in G$, introduce a formal symbol z_g . The group algebra $\mathbb{K}[G]$ consists of formal sums

$$\sum_{g \in G} a_g z_g, \quad a_g \in \mathbb{K},$$

in which only finitely many coefficients are nonzero. Here z_g are formal symbols. We add by collecting coefficients of the same symbol, and multiply by distributing and using the rule

$$z_g z_h = z_{gh}.$$

Thus multiplication of the symbols records multiplication in the group. Two sums are equal exactly when the coefficient of each symbol agrees. The multiplicative identity is z_e , where e is the identity of G . We define $\mathbb{Z}[G]$ analogously.

- *The Weyl algebra.* Suppose $\mathbb{K}$ has characteristic zero, meaning that $n \cdot 1_{\mathbb{K}} \neq 0$ for every positive integer n . Let $\text{End}_{\mathbb{K}}(\mathbb{K}[t])$ denote the ring of $\mathbb{K}$ -linear maps $\mathbb{K}[t] \rightarrow \mathbb{K}[t]$, where addition is pointwise and multiplication is composition. Consider the following two $\mathbb{K}$ -linear endomorphisms

$$\begin{array}{ccc} \mathbb{K}[t] & \xrightarrow{X} & \mathbb{K}[t] \\ f & \longmapsto & tf \end{array} \qquad \begin{array}{ccc} \mathbb{K}[t] & \xrightarrow{\partial} & \mathbb{K}[t] \\ f & \longmapsto & f' \end{array}$$

where f' is the formal derivative of f . The first Weyl algebra $A_1(\mathbb{K})$ is the subring of $\text{End}_{\mathbb{K}}(\mathbb{K}[t])$ generated by X, ∂ , and the scalar multiples $c \cdot \text{Id}$ of the identity operator. That is, elements of $A_1(\mathbb{K})$ are the finite sums of composites of copies of X and ∂ , scaled by elements of $\mathbb{K}$.

Since the product is composition, the operator on the right acts first. So ∂X means “multiply by t , then differentiate,” while $X\partial$ means “differentiate, then multiply by t .” By the product rule these differ:

$$(\partial X)(f) = (tf)' = f + tf', \quad (X\partial)(f) = tf'.$$

Subtracting, $(\partial X - X\partial)(f) = f$ for every $f \in \mathbb{K}[t]$, that is $\partial X - X\partial = \text{Id}$. Since $\text{Id} \neq 0$, we get $\partial X \neq X\partial$, so $A_1(\mathbb{K})$ is not commutative.

Problem 12. Power Series, Endomorphisms, and Other Ring Constructions.

- (a) In $\mathbb{K}[[t]]$, prove that $f = \sum a_j t^j$ can be a unit only if $a_0 \neq 0$. When $a_0 \neq 0$, look for $g = \sum b_j t^j$ with $fg = 1$ by solving first for b_0 and then recursively for b_n . Use this to characterize $\mathbb{K}[[t]]^\times$. Compare the inverse of $1 - t$ here with the

behavior of $1 - t$ in $\mathbb{K}[t]$, which you determined in Problem .

- (b) Check that pointwise addition and composition make $\text{End}_{\mathbb{Z}}(B)$ a ring.
- (c) Verify that the upper-triangular matrices form a subring of $\text{Mat}_2(\mathbb{K})$ containing the identity. Exhibit two of them which do not commute.
- (d) In $\mathbb{K}[S_3]$, compute $((12) + (23))^2$, using right-to-left composition of permutations. Explain why this ring is noncommutative over every field, including in characteristic two.
- (e) Evaluate $(\partial X - X\partial)(t^j)$ for every $j \geq 0$ and deduce that $\partial X - X\partial = 1$ in $A_1(\mathbb{K})$. Explain why the 1 here denotes an identity operator rather than a constant polynomial.
- (f) Determine the units of the ring of upper-triangular matrices in $\text{Mat}_2(\mathbb{K})$, and show that this ring has nonzero nilpotent elements. Compare with your answer for the dual numbers in Problem .

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)

Worksheet 1.2: Modules, Submodules, and Quotients

Let R be a ring. A *left R -module* is an abelian group $(M, +, 0_M)$ together with a scalar multiplication $R \times M \rightarrow M$, written $(r, m) \mapsto rm$, satisfying the following axioms:

- (i) (Identity) $1_R m = m$ for all $m \in M$,
- (ii) (Associativity) $(rs)m = r(sm)$ for all $r, s \in R$ and all $m \in M$,
- (iii) (Left Distributivity) $r(m + n) = rm + rn$ for all $r \in R$ and all $m, n \in M$, and
- (iv) (Right Distributivity) $(r + s)m = rm + sm$ for all $r, s \in R$ and all $m \in M$.

We write rm for the scalar multiplication of $r \in R$ and $m \in M$, and we write 0 for 0_R or 0_M when the intended identity is clear from context. Unlike a ring, a module need not have a multiplication of its own elements. The associativity axiom describes how multiplication in R interacts with its action on M : acting by s and then by r is the same as acting by rs .

A *right R -module* has its scalars acting on the right, with associativity given by $m(rs) = (mr)s$. Unless we say otherwise, *module* will always mean left module. In a commutative ring the two notions agree by setting $mr = rm$, but in an arbitrary ring it is important to keep track of the side on which scalars act. As in the previous worksheet all rings have a multiplicative identity, but need not be commutative, and the letter $\mathbb{K}$ will denote a field.

The goal of this worksheet is to introduce modules and study their submodules, homomorphisms, and quotients. Modules generalize vector spaces by allowing scalars to come from a ring. You already know several examples, including vector spaces and abelian groups. The definitions let us study these examples together, while retaining differences such as whether a module has a basis.

Problem 1. Basic Module Identities. Throughout this exercise let M be a left R -module.

- (a) Starting from the axioms, prove the following identities:
 - (i) $r0_M = 0_M$ for all $r \in R$,
 - (ii) $0_R m = 0_M$ for all $m \in M$, and
 - (iii) $(-r)m = -(rm) = r(-m)$ for all $r \in R$ and $m \in M$.
- (b) State in words what each of the three identities above mean.

Solution.

(a) (i)

(ii)

(iii)

(b)

Problem 2. First Examples of Modules. Throughout this exercise let R be a ring.

- (a) Compare the module axioms with the vector space axioms, and explain why a $\mathbb{K}$ -module is exactly a $\mathbb{K}$ -vector space. Do the module axioms require nonzero scalars to be invertible? Do they require multiplication in R to be commutative?
- (b) Check that R itself is a left R -module under its own multiplication. Which module axioms are inherited from which ring axioms?
- (c) If $I \subset R$ is a left ideal, explain why the multiplication on R makes I an R -module.
- (d) Given an example of a ring R and an additive subgroup $A \subset R$, which is not an R -module.
- (e) Let A be an abelian group. Define an action $\mathbb{Z} \times A \rightarrow A$ by $(n, a) \mapsto na$ where na means a added to itself n times. Show that this makes A into a $\mathbb{Z}$ -module.
- (f) Conversely, show that the axioms force every $\mathbb{Z}$ -action on A to be this one. Thus an abelian group has a unique $\mathbb{Z}$ -module structure. Compare this with the unique ring homomorphism $\mathbb{Z} \rightarrow R$ from Worksheet 1.1.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)

Let M be an R -module. A *submodule* $N \subset M$ is an additive subgroup which is closed under the R -action, so that the action of R on M restrict to an action of R on N . (There might be some sort of diagram you might draw here.) Equivalently, N is nonempty and $rn + sn' \in N$ whenever $r, s \in R$ and $n, n' \in N$. This is the same closure condition used to recognize subspaces, with the scalars now taken from R .

Let M and N be R -modules, a *module homomorphism*, or *R -linear map*, is a function $\phi : M \rightarrow N$ satisfying the following conditions for all $m, m' \in M$ and all $r \in R$:

- (i) $\phi(m + m') = \phi(m) + \phi(m')$.
- (ii) $\phi(rm) = r\phi(m)$.

Notice the first condition says that ϕ is a homomorphism of the underlying abelian groups. In particular it preserves 0 and additive inverses. The second says that ϕ respects the scalar action. We write $\text{Hom}_R(M, N)$ for the set of such maps. We call a homomorphism from a module to itself a *endomorphism* and so $\text{End}_R(M) := \text{Hom}_R(M, M)$.

A module homomorphism is an *isomorphism* if it has an inverse which is again a module homomorphism. As with groups and rings, this turns out to be equivalent to the set theoretic condition of being a bijection. We write $M \cong N$ when an isomorphism exists. An *automorphism* of M is an isomorphism $\phi : M \rightarrow M$. We denote the set of R -linear automorphisms by $\text{Aut}_R(M) \subset \text{End}_R(M)$.

As with ring homomorphisms, a module homomorphism $\phi : M \rightarrow N$ comes with a *kernel* and an *image*:

$$\ker(\phi) := \{m \in M \mid \phi(m) = 0_N\} \subset M \quad \text{and} \quad \text{img}(\phi) := \{\phi(m) \mid m \in M\} \subset N.$$

A homomorphism is surjective exactly when its image is the whole target, while its kernel measures its failure to be injective. Whether an additive subgroup is a submodule depends on the ring action. For example, a subspace of $\mathbb{K}^2$ need not be preserved by every matrix in $\text{Mat}_2(\mathbb{K})$. We will see this distinction in the following exercise.

Problem 3. Submodules and Module Homomorphisms. Throughout this exercise let M and N be R -modules.

- (a) Prove that $N \subset M$ is a non-empty R -submodule if and only if N is non-empty and $rn + sn' \in N$ whenever $r, s \in R$ and $n, n' \in N$.
- (b) Show that $\mathbb{K}$ -module maps are exactly the linear maps of vector spaces.
- (c) Show that $\mathbb{Z}$ -module maps are exactly the homomorphisms of abelian groups.
- (d) Let $\phi : M \rightarrow N$ be an R -linear map. Prove that $\ker(\phi)$ is a submodule of M and that $\text{img}(\phi)$ is a submodule of N .

- (e) Prove that ϕ is injective if and only if $\ker(\phi) = \{0\}$, and surjective if and only if $\text{img}(\phi) = P$.
- (f) Show that compositions of R -linear maps and identity maps are R -linear. If ϕ is a bijective R -linear map, prove that ϕ^{-1} is R -linear, so that ϕ is an isomorphism. Hint: write two elements of the target as $\phi(m)$ and $\phi(m')$.
- (g) Let $R = \text{Mat}_2(\mathbb{K})$ act on the column space $\mathbb{K}^2$ by matrix multiplication. Verify the module axioms. Show that any nonzero column can be carried to any other column by a suitable matrix, and deduce that this module has exactly two submodules. In particular, is the line $\mathbb{K}e_1$ an R -submodule? Compare this with the left ideal of $\text{Mat}_2(\mathbb{K})$ consisting of matrices whose second column is zero, which you studied in Worksheet 1.1.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)
- (g)

Modules over the polynomial ring $\mathbb{K}[x]$ give a useful dictionary between module theory and linear algebra. Let V be a $\mathbb{K}$ -vector space and let $T : V \rightarrow V$ be a linear operator. Substituting T for x defines an action of $\mathbb{K}[x]$ on V by the rule

$$(a_0 + a_1x + \cdots + a_dx^d)v := a_0v + a_1T(v) + \cdots + a_dT^d(v), \quad T^0 := \text{Id}_V.$$

We write V_T for the resulting module. Constant polynomials act by the original scalar multiplication on V , and the variable x acts by T . Conversely, a $\mathbb{K}[x]$ -module becomes a vector space once we restrict the scalars to the constant polynomials, and multiplication by x is then a linear operator on that vector space. A $\mathbb{K}[x]$ -module is therefore the same thing as a vector space together with a choice of linear operator. The variable x is still a formal symbol; it is the module structure which specifies how it acts.

This dictionary also translates submodules and homomorphisms. Let W be a $\mathbb{K}$ -vector space with a linear operator $S : W \rightarrow W$. A $\mathbb{K}$ -linear map $\phi : V \rightarrow W$ is a module homomorphism $V_T \rightarrow W_S$ if and only if $\phi \circ T = S \circ \phi$, in which case we call ϕ an *intertwining map*. Similarly, a subspace $U \subset V$ is a submodule of V_T if and only if $T(U) \subset U$, in which case we call U a *T -invariant subspace*. These translations let us study a linear operator by studying its module, a point of view we will return to when we discuss canonical forms. We check the dictionary in the following exercise.

Problem 4. Modules over $\mathbb{K}[x]$ and Linear Operators. Throughout this exercise let V and W be $\mathbb{K}$ -vector spaces equipped with linear operators T and S respectively.

- (a) Verify that the displayed formula makes V into a $\mathbb{K}[x]$ -module. For the associativity axiom, first check that $T^iT^j = T^{i+j}$, and then expand a product of two polynomials.
- (b) Conversely, let M be a $\mathbb{K}[x]$ -module. Use the constant polynomials to regard M as a $\mathbb{K}$ -vector space, and set $T(m) := xm$. Prove that T is $\mathbb{K}$ -linear, and then show that every polynomial acts by the displayed formula. Conclude that the two constructions undo one another.
- (c) Prove that a map $\phi : V \rightarrow W$ is $\mathbb{K}[x]$ -linear as a map $V_T \rightarrow W_S$ if and only if it is $\mathbb{K}$ -linear and

$$\phi \circ T = S \circ \phi.$$

For the reverse implication, first prove that $\phi \circ T^j = S^j \circ \phi$ by induction on j .

- (d) Prove that the $\mathbb{K}[x]$ -submodules of V_T are exactly the subspaces $U \subset V$ with $T(U) \subset U$. Why does stability under T already force stability under every polynomial in T ?

- (e) Show that an intertwining map carries T -invariant subspaces to S -invariant subspaces. Check directly that its kernel is T -invariant and that its image is S -invariant.
- (f) Suppose that V and W are finite dimensional. Choose bases and translate the equation $\phi \circ T = S \circ \phi$ into an equation of matrices. Deduce that $V_T \cong W_S$ exactly when the matrices of the two operators are related by an invertible change of coordinates.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)

For a concrete example, let $V = \mathbb{K}^2$ with basis e_1, e_2 , and define a linear operator by $T(e_1) = 0$ and $T(e_2) = e_1$. The operator T is *nilpotent*, meaning that some positive power of T is zero. Here $T^2 = 0$, so a polynomial $f(x) = a_0 + a_1x + x^2g(x)$ acts by

$$f(T)(ue_1 + ve_2) = (a_0u + a_1v)e_1 + a_0ve_2.$$

In particular, two polynomials which differ by a multiple of x^2 act in the same way. Recall the ring of dual numbers $\mathbb{K}[\varepsilon] = \mathbb{K}[x]/\langle x^2 \rangle$ from Worksheet 1.1, where $\varepsilon = x + \langle x^2 \rangle$. Every element has a unique expression $a + b\varepsilon$ with $a, b \in \mathbb{K}$, and $\varepsilon^2 = 0$. Multiplication by ε sends 1 to ε and ε to zero, just as T sends e_2 to e_1 and e_1 to zero. We use this comparison in the following exercise to identify the two modules, and then to compute their submodules and endomorphisms.

Problem 5. A Nilpotent Operator and the Dual Numbers. Throughout this exercise let V and T be as above, let $\mathbb{K}[\varepsilon] = \mathbb{K}[x]/\langle x^2 \rangle$, and let $\varepsilon = x + \langle x^2 \rangle$.

- (a) Use the displayed formula to compute $(2 + 3x + x^5)e_2$ and $x(e_1 + e_2)$ in V_T . Compute these vectors when $\mathbb{K} = \mathcal{F}_2$.
- (b) Regard $\mathbb{K}[\varepsilon]$ as a $\mathbb{K}[x]$ -module by $f(x) \cdot (a + b\varepsilon) := f(\varepsilon)(a + b\varepsilon)$. Show directly that

$$\phi : \mathbb{K}[\varepsilon] \longrightarrow V_T, \quad a + b\varepsilon \longmapsto ae_2 + be_1$$

is well defined, $\mathbb{K}[x]$ -linear, and bijective. Explain why interchanging e_1 and e_2 in this formula would not give a module homomorphism.

- (c) Show that e_2 generates V_T as a $\mathbb{K}[x]$ -module, in the sense that every vector is of the form $f(x)e_2$. Does e_2 alone span V as a $\mathbb{K}$ -vector space? Explain why the two questions have different answers.
- (d) Compute $\ker(T)$ and $\text{img}(T)$. Use Problem 5 to show that V_T has exactly three submodules: $\{0\}$, $\mathbb{K}e_1$, and V_T . Hint: an invariant subspace containing $ue_1 + ve_2$ with $v \neq 0$ also contains e_1 .
- (e) Write an arbitrary $\mathbb{K}$ -linear map $\phi : V \rightarrow V$ as a 2×2 matrix and solve the equation $\phi \circ T = T \circ \phi$. Describe every element of $\text{End}_{\mathbb{K}[x]}(V_T)$, and determine which of them lie in $\text{Aut}_{\mathbb{K}[x]}(V_T)$. Compare the automorphisms you find with the units of $\mathbb{K}[\varepsilon]$ from Worksheet 1.1.

Solution.

(a)

(b)

(c)

(d)

(e)

Let $N \subset M$ be a submodule. Since N is an additive subgroup of the abelian group M , we may form the quotient group M/N , whose elements are the cosets $m + N$, with $m + N = m' + N$ if and only if $m - m' \in N$. To obtain a *quotient module*, we define a scalar multiplication by

$$r(m + N) := rm + N.$$

The fact that N is closed under the action of R guarantees that this formula does not depend on the chosen representative. Together with the addition $(m + N) + (m' + N) = (m + m') + N$, it makes M/N an R -module. The *quotient map*

$$M \xrightarrow{\pi} M/N \qquad m \longmapsto m + N$$

is a surjective module homomorphism with kernel N . Consequently, every submodule occurs as the kernel of a module homomorphism, just as every two-sided ideal occurs as the kernel of a ring homomorphism.

One way to think of M/N is as the module obtained by imposing the equations $n = 0$ for all $n \in N$. More precisely, a module homomorphism $\phi : M \rightarrow P$ which vanishes on N determines a unique module homomorphism $\bar{\phi} : M/N \rightarrow P$ such that $\bar{\phi} \circ \pi = \phi$:

$$\begin{array}{ccc} M & \xrightarrow{\phi} & P \\ \pi \downarrow & \nearrow \bar{\phi} & \\ M/N & & \end{array}$$

This is the *universal property of quotient modules*. As with quotient rings, it describes homomorphisms out of M/N in terms of homomorphisms out of M which send N to zero. We check the construction and this property in the following exercise.

Problem 6. Construction and Properties of Quotient Modules. Throughout this exercise let $N \subset M$ be a submodule of an R -module.

- (a) Suppose $m + N = m' + N$, and prove that $rm + N = rm' + N$ for every $r \in R$, using the fact that N is closed under scalar multiplication. Write down the formulas for addition and scalar multiplication in M/N , and verify the module axioms.
- (b) Prove that $\pi : M \rightarrow M/N, m \mapsto m + N$, is a surjective module homomorphism with $\ker(\pi) = N$. Conclude that every submodule occurs as a kernel.
- (c) Suppose $\phi : M \rightarrow P$ is R -linear with $N \subset \ker(\phi)$. Prove that $\bar{\phi}(m + N) = \phi(m)$ is a well-defined R -linear map, and that it is the unique map $M/N \rightarrow P$

satisfying $\bar{\phi} \circ \pi = \phi$. Explain in words why a homomorphism out of M/N is the same thing as a homomorphism out of M which sends N to zero.

Solution.

- (a)
- (b)
- (c)

The construction of quotient modules also explains the relationship between the kernel and the image of a homomorphism $\phi : M \rightarrow P$. Two elements of M have the same image exactly when their difference lies in $\ker(\phi)$. Passing to $M / \ker(\phi)$ therefore identifies exactly the elements that ϕ identifies. The induced map is injective and has image $\text{img}(\phi)$.

Theorem 0.2 (First Isomorphism Theorem for Modules). *Let $\phi : M \rightarrow P$ be a homomorphism of R -modules. Then the map*

$$\bar{\phi} : M / \ker(\phi) \longrightarrow \text{img}(\phi), \quad m + \ker(\phi) \longmapsto \phi(m),$$

is an isomorphism of R -modules. In particular, there is a commutative diagram

$$\begin{array}{ccc} M & \xrightarrow{\phi} & P \\ \pi \downarrow & & \uparrow \\ M / \ker(\phi) & \xrightarrow[\sim]{\bar{\phi}} & \text{img}(\phi) \end{array}$$

Every module homomorphism can therefore be expressed as a quotient map followed by an isomorphism and an inclusion. If ϕ is surjective, the theorem gives $M / \ker(\phi) \cong P$. As with rings, we can recognize a quotient module by constructing a surjective homomorphism onto a module we already understand and computing its kernel. The familiar versions for abelian groups and vector spaces are special cases. We prove the theorem and apply it in the following exercise.

Problem 7. The First Isomorphism Theorem and Examples. Throughout parts (a) and (b) let $\phi : M \rightarrow P$ be a homomorphism of R -modules.

- (a) Use Problem with $N = \ker(\phi)$ to construct $\bar{\phi} : M / \ker(\phi) \rightarrow \text{img}(\phi)$. Show that the resulting map is surjective, and that if $\bar{\phi}(m + \ker(\phi)) = 0$ then $m + \ker(\phi)$ is the zero coset. Deduce that $\bar{\phi}$ is injective.
- (b) Use Problem to conclude that the bijection of part (a) is an isomorphism of R -modules, which proves the theorem.
- (c) Apply the theorem to the $\mathbb{Z}$ -linear map $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/12\mathbb{Z}, a \mapsto \overline{8a}$. Compute its kernel and its image, and write the induced isomorphism down explicitly. What does it do to the class of 1?
- (d) Return to the module V_T of Problem . Regard T itself as a $\mathbb{K}[x]$ -linear map and identify $V_T / \ker(T) \cong \text{img}(T)$, describing the action of x on each side. For an

invariant subspace U of an arbitrary V_T , explain why the operator induced on the quotient is $v + U \mapsto T(v) + U$.

Solution.

- (a)
- (b)
- (c)
- (d)

The first isomorphism theorem also lets us compare quotients which are constructed in different ways. Let $A, B \subset M$ be submodules. Their *sum* is

$$A + B := \{a + b \mid a \in A, b \in B\},$$

which is the smallest submodule of M containing both A and B . Passing to $(A + B)/B$ has the effect of setting the elements of B equal to zero, so every class can be represented by an element of A alone. The part of A which becomes zero in the process is exactly $A \cap B$. The second isomorphism theorem is the precise form of this observation, and its proof is a single application of the first.

Theorem 0.3 (Second Isomorphism Theorem for Modules). *If A and B are submodules of an R -module M , then*

$$A/(A \cap B) \xrightarrow{\sim} (A + B)/B, \quad a + (A \cap B) \mapsto a + B$$

is an isomorphism of R -modules.

Problem 8. The Second Isomorphism Theorem. Throughout this exercise let A and B be submodules of an R -module M .

- (a) Prove that $A + B$ and $A \cap B$ are submodules of M , and that $A + B$ is contained in every submodule of M containing both A and B .
- (b) Define $\phi : A \rightarrow (A + B)/B$ by $a \mapsto a + B$, and verify that it is a module homomorphism. Why does its image contain the class of every element $a + b$ of $A + B$?
- (c) Compute $\ker(\phi)$, and use Problem 1 to obtain the isomorphism

$$A/(A \cap B) \cong (A + B)/B.$$

Write down the image of a general coset, and check directly that your formula does not depend on the chosen representative.

- (d) Take $M = \mathbb{Z}$ as a $\mathbb{Z}$ -module, with $A = 4\mathbb{Z}$ and $B = 6\mathbb{Z}$. Compute all four of the submodules appearing in the theorem, list the elements of both quotients, and match them under your isomorphism. Identify an element generating each quotient, and say where it goes.

Solution.

(a)

(b)

(c)

(d)

Suppose $N \subset L \subset M$ are submodules. We may form M/L directly, or we may first form M/N and then take the quotient by L/N . The third isomorphism theorem says that these two constructions give isomorphic modules. The correspondence theorem describes all submodules of M/N in terms of the submodules of M which contain N .

Theorem 0.4 (Third Isomorphism and Correspondence Theorems). *Let $N \subset M$ be a submodule and let $\pi : M \rightarrow M/N$ be the quotient map. The assignments $L \mapsto L/N = \pi(L)$ and $U \mapsto \pi^{-1}(U)$ give mutually inverse, inclusion-preserving bijections*

$$\{L \subset M \mid L \text{ a submodule containing } N\} \longleftrightarrow \{U \subset M/N \mid U \text{ a submodule}\}.$$

Moreover, for $N \subset L \subset M$ there is an isomorphism of R -modules

$$(M/N)/(L/N) \xrightarrow{\sim} M/L, \quad (m + N) + (L/N) \mapsto m + L.$$

Since the quotient map sends every element of N to zero, the preimage of any submodule of M/N contains N . Conversely, a submodule $L \subset M$ containing N determines the submodule $L/N \subset M/N$. The correspondence theorem says that these two constructions undo one another and preserve inclusion. We prove both statements in the following exercise.

Problem 9. The Third Isomorphism and Correspondence Theorems. Throughout this exercise let $N \subset L \subset M$ be submodules and let $\pi : M \rightarrow M/N$ be the quotient map.

- (a) Show that $L/N = \{\ell + N \mid \ell \in L\}$ is a submodule of M/N , and that $\phi : M/N \rightarrow M/L, m + N \mapsto m + L$, is well defined and R -linear. Where do you use the inclusion $N \subset L$?
- (b) Show that ϕ is surjective with $\ker(\phi) = L/N$. Deduce the third isomorphism theorem, and write its formula on a coset of a coset.
- (c) If U is a submodule of M/N , prove that $\pi^{-1}(U)$ is a submodule of M containing N .
- (d) Prove the two identities

$$\pi^{-1}(L/N) = L, \quad \pi(\pi^{-1}(U)) = U,$$

and conclude that $L \mapsto L/N$ and $U \mapsto \pi^{-1}(U)$ are inverse bijections. This is the correspondence theorem.

- (e) Check that both assignments preserve inclusion. What is $\pi^{-1}(0)$, and what is $\pi^{-1}(M/N)$? Explain why the submodules of M which do not contain N are excluded.

- (f) Take $M = \mathbb{Z}$ and $N = 12\mathbb{Z}$. List every submodule of $\mathbb{Z}$ containing $12\mathbb{Z}$, together with its corresponding submodule of $\mathbb{Z}/12\mathbb{Z}$. Then take $L = 4\mathbb{Z}$ and write out the third isomorphism theorem for these particular modules.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)

Let M and P be R -modules. The set $\text{Hom}_R(M, P)$ is itself an abelian group under pointwise addition:

$$(f + g)(m) = f(m) + g(m), \quad (-f)(m) = -f(m),$$

with the zero homomorphism as its identity element. Thus the collection of homomorphisms between two modules is again an algebraic object of the same general kind. Over a commutative ring, pointwise scalar multiplication also makes $\text{Hom}_R(M, P)$ an R -module. Since our rings need not be commutative, we use only its abelian group structure here.

Maps out of cyclic modules give the first computations of this kind. Such a map is determined by the image of a generator, but that image is not arbitrary: it must satisfy whatever relations the generator satisfies. For a positive integer m , a homomorphism from $\mathbb{Z}/m\mathbb{Z}$ to an abelian group P is therefore the same thing as an element $y \in P$ with $my = 0$. When $P = \mathbb{Z}/n\mathbb{Z}$ this description leads to

$$\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/\gcd(m, n)\mathbb{Z}.$$

In the following exercise we construct this isomorphism by finding the possible images of $\bar{1}$, and then use the same description to compute kernels and images in some examples.

Problem 10. Homomorphisms between Cyclic Abelian Groups. Throughout parts (b) through (g) let $m, n \geq 1$ be integers and put $d := \gcd(m, n)$.

- (a) Let M and P be R -modules. Check that pointwise addition and negation preserve R -linearity. Identify the zero map, and verify that $\text{Hom}_R(M, P)$ is an abelian group.
- (b) Show that a homomorphism $\phi : \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is determined by $y = \phi(\bar{1})$, and prove that $my = 0$ is necessary. Conversely, if $my = 0$, show that $\phi(\bar{a}) = ay$ is well defined and a homomorphism.
- (c) Write $m = dm'$ and $n = dn'$ with $\gcd(m', n') = 1$. Show that $my = 0$ in $\mathbb{Z}/n\mathbb{Z}$ exactly when y is represented by a multiple of n' . Hint: use Bézout's identity to justify the divisibility step.
- (d) Show that the assignment

$$\mathbb{Z}/d\mathbb{Z} \longrightarrow \text{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}), \quad \bar{t} \longmapsto (\bar{a} \longmapsto \overline{a(n/d)t})$$

is a well-defined isomorphism of abelian groups. In particular, exactly how many such homomorphisms are there?

- (e) List every map $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/8\mathbb{Z}$ by its value on $\bar{1}$, and compute the kernel and the image of each. Check Problem against these examples, both numerically and by writing down the induced isomorphism.

- (f) Consider $\phi : \mathbb{Z}/12\mathbb{Z} \rightarrow \mathbb{Z}/18\mathbb{Z}$ with $\phi(\bar{1}) = \bar{6}$. Verify that it exists, and compute its kernel, its image, and the induced isomorphism on the quotient. Does a map with $\phi(\bar{1}) = \bar{1}$ exist? Explain using the relation satisfied by $\bar{1}$ in the domain.
- (g) Use the possible orders of $\phi(\bar{1})$ to show that an injection $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ exists exactly when $m \mid n$, while a surjection exists exactly when $n \mid m$. Write down explicit maps in both cases.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)
- (g)

We finish by describing module structures in terms of ring homomorphisms. Recall from Worksheet 1.1 that for an abelian group M , the endomorphism ring $\text{End}_{\mathbb{Z}}(M) = \text{Hom}_{\mathbb{Z}}(M, M)$ has pointwise addition and multiplication given by composition, with $fg = f \circ g$. Its multiplicative identity is Id_M .

If M is an R -module, then multiplication by a scalar r is an additive map $\rho(r) : M \rightarrow M$, where $\rho(r)(m) = rm$. The module axioms say exactly that the resulting function

$$\rho : R \longrightarrow \text{End}_{\mathbb{Z}}(M)$$

is a ring homomorphism. For instance, the associativity axiom $(rs)m = r(sm)$ becomes the identity $\rho(rs) = \rho(r) \circ \rho(s)$, and the identity axiom $1_R m = m$ becomes $\rho(1_R) = \text{Id}_M$, which is condition (iii) in the definition of a ring homomorphism. Conversely, any such ring homomorphism defines an R -module structure on the abelian group M . Thus giving an R -module structure on M is the same thing as giving a ring homomorphism $R \rightarrow \text{End}_{\mathbb{Z}}(M)$.

Once an R -action has been fixed we may form the ring $\text{End}_R(M)$ of those endomorphisms which respect it. This is a subring of $\text{End}_{\mathbb{Z}}(M)$, with the same addition, composition, and identity element, and it acts on M by *evaluation*, $f \cdot m = f(m)$. The condition $f(rm) = rf(m)$ says precisely that this action commutes with the original one. Note that over a noncommutative ring the scalar operators $\rho(r)$ need not themselves be R -linear, which is why the target of ρ is $\text{End}_{\mathbb{Z}}(M)$ and not $\text{End}_R(M)$. The column module over $\text{Mat}_2(\mathbb{K})$ makes this distinction concrete, as we see in the following exercise.

Problem 11. Module Structures and Endomorphism Rings. Throughout this exercise let M be an R -module.

- (a) Check the ring axioms for $\text{End}_{\mathbb{Z}}(M)$. For one of the two distributive laws you will need that the operators are additive; identify which one, and explain why the other is formal. Show that $\text{End}_R(M)$ is a subring containing the same identity element.
- (b) Starting from the module structure, define

$$\rho : R \rightarrow \text{End}_{\mathbb{Z}}(M), \quad \rho(r)(m) = rm.$$

Show that each $\rho(r)$ is additive and that ρ is a ring homomorphism, and identify which module axiom gives each property.

- (c) Conversely, given a ring homomorphism $\rho : R \rightarrow \text{End}_{\mathbb{Z}}(M)$, set $rm := \rho(r)(m)$ and verify the module axioms. Show that this construction inverts the one in part (b). Describe $\rho(x)$ for a $\mathbb{K}[x]$ -module V_T .
- (d) Prove that M is a left $\text{End}_R(M)$ -module by evaluation, $f \cdot m = f(m)$, using the convention $fg = f \circ g$. Show also that the condition $f(rm) = rf(m)$ says that these operators commute with the original scalar operators.

- (e) For the column module $M = \mathbb{K}^2$ over $R = \text{Mat}_2(\mathbb{K})$, show that ρ is injective. Show also that $\rho(A)$ need not be R -linear, by taking $A = E_{12}$ and $B = E_{21}$ and comparing $A(Be_1)$ with $B(Ae_1)$; here E_{ij} denotes the matrix units of Worksheet 1.1.
- (f) Determine $\text{End}_{\text{Mat}_2(\mathbb{K})}(\mathbb{K}^2)$. First use the scalar matrices to show that every module endomorphism is $\mathbb{K}$ -linear, and then require its matrix to commute with E_{11} , E_{12} , and E_{21} . Compare the answer with the endomorphism ring you computed in Problem , and say what each computation tells you about the module.

Solution.

- (a)
- (b)
- (c)
- (d)
- (e)
- (f)